

Existence

Existence of Walrasian Equilibria

Definition 1 (Simplex) The simplex in $\mathbb{R}^L$ is the set

$$\Delta = \left\{ p \in \mathbb{R}_+^L : \sum_{l=1}^L p_l = 1 \right\}.$$

- Its relative interior is $\Delta^\circ = \{p \in \Delta : p \gg 0\}$.

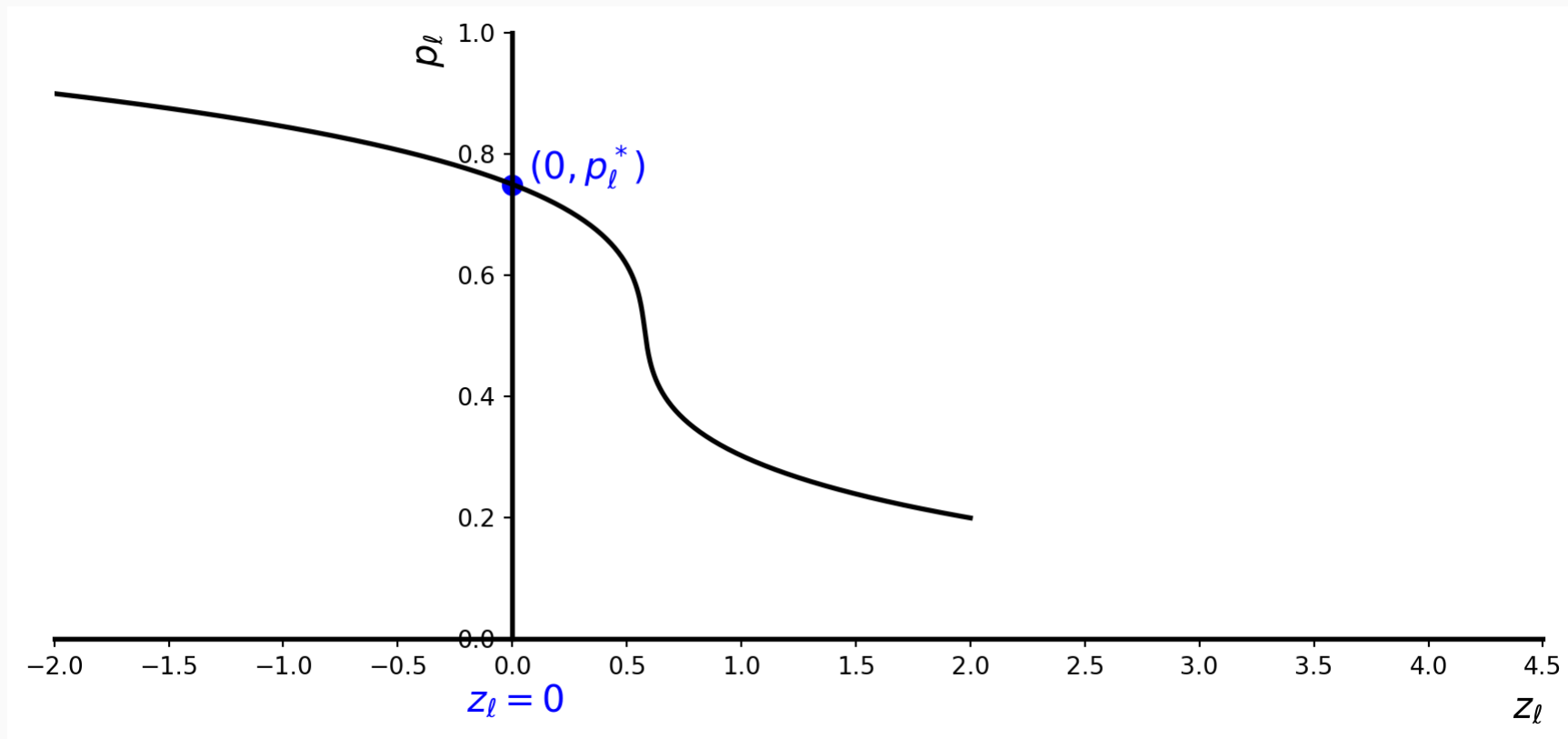
Definition 2 (Aggregate excess demand) Let $\mathcal{E}$ be an exchange economy. Its aggregate excess demand is the correspondence $z : \Delta^\circ \rightarrow \mathbb{R}^L$,

$$z(p) = \sum_{i=1}^I (D_i(p, \omega_i) - \omega_i).$$

- If a WE does not exist under reasonable assumptions, there is no *theory* or *application*.

Intuition

$$z(p) = z(p_1, \dots, p_L) = (z_1(p), \dots, \underbrace{z_\ell(p), \dots, z_L(p)}_{z_\ell(p_1, \dots, p_L)})$$



Debreu-Gale-Kuhn-Nikaido lemma

Lemma 1 (Debreu-Gale-Kuhn-Nikaido) Let $z : \Delta^\circ \rightarrow \mathbb{R}^L$ be a function satisfying

1. Continuity
2. Walras' Law: $(\forall p \in \Delta^\circ) p \cdot z(p) = 0$
3. Boundedness below: $(\exists x \in \mathbb{R}^L)(\forall p \in \Delta^\circ) z(p) \geq x$
4. Boundary Condition:

$$[p^n \rightarrow p, p \in \Delta \setminus \Delta^\circ] \implies |z(p^n)| \rightarrow \infty.$$

Then $\exists p^* \in \Delta^\circ$ such that $z(p^*) = 0$.

Walras' Law

- Suppose agents spend their entire income:

$$(\forall p \in \Delta^\circ)(\forall i)[\forall x_i \in D_i(p, \omega_i)], p \cdot x_i = p \cdot \omega_i$$

For an agent i , this follows from LNS or monotonicity of $\succeq_i$.

- Aggregating over consumers:

$$[\forall i, x_i \in D_i(p, \omega_i)] \implies \sum_{i=1}^I p \cdot x_i = \sum_{i=1}^I p \cdot \omega_i$$

Thus

$$\begin{aligned} 0 &= \sum_{i=1}^I p \cdot x_i - \sum_{i=1}^I p \cdot \omega_i \\ &= p \cdot \sum_{i=1}^I (x_i - \omega_i) \\ &= p \cdot \sum_{i=1}^I [D_i(p, \omega_i) - \omega_i] \\ &= p \cdot z(p) \end{aligned}$$

Walras' Law

- Suppose $p \gg 0$ and $p \cdot z(p) = 0$.
- To show that p clears all markets, it suffices to show that it clears all but one market:
- Suppose $\ell = 2, \dots, L$, $z_\ell(p) = 0$.
- Since $0 = p \cdot z(p) = p_1 \cdot z_1(p) + \sum_{\ell=2}^L p_\ell \cdot z_\ell(p)$
- It follows $p_1 \cdot z_1(p) = 0$ and therefore $z_1(p) = 0$.

Outline of proof DGKN

- Define a correspondence $\varphi : \Delta^\circ \rightarrow \Delta$ by $\varphi(p) = \{q \in \Delta : q \cdot z(p) \geq q' \cdot z(p), \forall q' \in \Delta\}$.

φ identifies the goods in highest excess demand.

- Extend the domain of φ to Δ to make it have closed graph.
- Verify that $p^* \in \varphi(p^*) \implies p^* \in \Delta^\circ \wedge z(p^*) = 0$.
- Check that φ satisfies the hypotheses of Kakutani's Theorem
- By Kakutani's Theorem, $\exists p^* \in \Delta : p^* \in \varphi(p^*)$.

Graph proof for $L = 2$

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Proof of DGKN lemma

1. Define the correspondence $\varphi : \Delta \rightarrow \Delta$ by

$$\varphi(p) = \begin{cases} \{q \in \Delta : (\forall q' \in \Delta) q \cdot z(p) \geq q' \cdot z(p)\} & \text{if } p \in \Delta^\circ \\ \{q \in \Delta : p \cdot q = 0\} & \text{if } p \in \Delta \setminus \Delta^\circ \end{cases}$$

- If $p \in \Delta^\circ$, $\varphi(p)$ picks prices that maximize the value of excess demand at p , that is to say, the goods with the largest excess demand at p , receive price 1.
- If $p \notin \Delta^\circ$, $\varphi(p)$ picks prices orthogonal to p .

2. Claim: $p^* \in \varphi(p^*) \implies p^* \in \Delta^\circ \wedge z(p^*) = 0$.

Proof. Suppose $p^* \in \varphi(p^*) \wedge p^* \in \Delta \setminus \Delta^\circ$.

Then,

$$\begin{aligned} & \forall q \in \varphi(p^*), p^* \cdot q = 0 \text{ (definition of } \varphi) \\ \implies & p^* \cdot p^* = 0 \text{ (because } p^* \in \varphi(p^*)) \\ \implies & p^* = 0 \wedge p^* \notin \Delta, \text{ a contradiction} \end{aligned}$$

This proves $p^* \in \Delta^\circ$.

Suppose $p^* \in \varphi(p^*) \wedge p^* \in \Delta^\circ$. Claim: $z(p^*) = 0$.

- If $z(p^*) < 0$ then $p^* \cdot z(p^*) < 0$, contradicting Walras' Law. Then $z(p^*) \not\leq 0$.
- Pick any $\ell \in \{1, \dots, L\}$ and let $q = (0, \dots, 0, \overset{\ell}{\widetilde{1}}, 0, \dots, 0) \in \Delta$.
- Then $z(p^*)_\ell = q_\ell z_\ell(p^*) = q \cdot z(p^*) \leq p^* \cdot z(p^*) = 0$ because $p^* \in \varphi(p^*)$ and Walras' Law. Conclude that $(\forall \ell) z_\ell(p^*) \leq 0$.
- Hence $z(p^*) \leq 0$ but $z(p^*) \not\leq 0$, so $z(p^*) = 0$.

This completes the proof of the claim.

3. Verify that φ satisfies the hypotheses of Kakutani's Fixed Point Theorem, i.e.

- Δ is compact, convex, a nonempty subset.
- $\varphi : \Delta \rightarrow \Delta$ is
 - a. nonempty valued,
 - b. convex valued, and
 - c. has a closed graph. See the closed graph theorem for correspondences.

3.a. φ is nonempty valued:

- Pick $p \in \Delta^\circ$. Then $q \cdot z(p)$ is a continuous function of $q \in \Delta$, a compact set. Weierstrass Theorem implies it achieves a maximum. Therefore $\varphi(p) \neq \emptyset$.
- Pick $p \in \Delta \setminus \Delta^\circ$. Then $p_\ell = 0$ for some ℓ . Let $q = (0, \dots, 0, \overset{\ell}{\underset{\sim}{1}}, 0, \dots, 0) \in \Delta$. Then $q \cdot p = 0$ and therefore $\varphi(p) \neq \emptyset$.

3.b. φ is convex-valued: Check.

3.c. φ has a closed graph, that is to say

If $p^n \rightarrow p, q^n \in \varphi(p^n), q^n \rightarrow q$, then $q \in \varphi(p)$.

- The proof considers three cases that depend on whether p is in the interior of the simplex or in its boundary, and on how it is approached.
1. $p \in \Delta^\circ$. (p is in the interior; it can only be approached from the interior.)
 2. $p \in \Delta \setminus \Delta^\circ$ and there is a subsequence $\{p^{n_k}\}$ of $\{p^n\}$ such that $(\forall k)$ $p^{n_k} \in \Delta^\circ$.
(p is in the boundary and it is approached from the interior.)
 3. $p \in \Delta \setminus \Delta^\circ$ and $(\forall n)$ $p^n \in \Delta \setminus \Delta^\circ$.
(p is in the boundary and it is approached from the boundary.)

Case 1. $p \in \Delta^\circ$.

- Then, $p^n \in \Delta^\circ$ for n sufficiently large.
- Since z is continuous on Δ° , then $z(p^n) \rightarrow z(p)$.

Pick any $q' \in \Delta$. Then

$$\begin{aligned} q' \cdot z(p) &= q' \cdot \lim_{n \rightarrow \infty} z(p^n) = \lim_{n \rightarrow \infty} q' \cdot z(p^n) \\ &\leq \lim_{n \rightarrow \infty} q^n \cdot z(p^n) \quad (\text{since } q^n \in \varphi(p^n)) \\ &= \lim_{n \rightarrow \infty} q^n \cdot \lim_{n \rightarrow \infty} z(p^n) = q \cdot z(p) \end{aligned}$$

Therefore $q \in \varphi(p)$.

Case 2. $p \in \Delta \setminus \Delta^\circ$ and there is a subsequence $\{p^{n_k}\}$ of $\{p^n\}$ such that $(\forall k) p^{n_k} \in \Delta^\circ$.

- Note that if $p \cdot q = 0$, then $q \in \varphi(p)$ by definition of φ . Therefore it suffices to show that $p \cdot q = 0$.
- Let $p_{\ell'} > 0$ and let $\alpha = \frac{p_{\ell'}}{2}$. Must show $q_{\ell'} = 0$.
- For k sufficiently large, $p_{\ell'}^{n_k} \geq \alpha$. (This bounds the price.)
- By Boundedness below, $(\exists x \in \mathbb{R}^L) (\forall n_k) z(p^{n_k}) \geq x$. Then

$$(\forall n_k)(\forall \ell) \left[-z_\ell(p^{n_k}) \leq -x_\ell \right]. \quad (1)$$

(This bounds the bundle.)

Using Equation 1 in the first inequality below,

$$\begin{aligned} p_{\ell'}^{n_k} z_{\ell'}(p^{n_k}) &= \overbrace{p^{n_k} \cdot z(p^{n_k})}^0 - \sum_{\ell \neq \ell'} p_{\ell}^{n_k} z_{\ell}(p^{n_k}) \\ &\leq - \sum_{\ell \neq \ell'} p_{\ell}^{n_k} x_{\ell} \\ &\leq \sum_{\ell \neq \ell'} p_{\ell}^{n_k} \|x\|_{\infty} \quad (\text{since } \forall \ell, |x_{\ell}| \leq \|x\|_{\infty}) \\ &\leq \|x\|_{\infty} \quad (\text{since } \sum_{\ell \neq \ell'} p_{\ell}^{n_k} \leq 1) \end{aligned}$$

- Then $z_{\ell'}(p^{n_k}) \leq \|x\|_{\infty} / \alpha$ since $p_{\ell'}^{n_k} > \alpha$.

- Since z satisfies the Boundary Condition and is bounded below, there is a sequence h^k with $h^k \in \{1, \dots, L\}$ such that $z_{h^k}(p^{n_k}) \rightarrow \infty$.
- Then for sufficiently large k , $z_{\ell'}(p^{n_k}) < z_{h^k}(p^{n_k})$.
- This implies $q_{\ell'}^{n_k} = 0$ and therefore $q_{\ell'} = 0$.
- Therefore $p_{\ell'} > 0 \implies q_{\ell'} = 0$.
- Then $q \cdot p = 0$ and $q \in \varphi(p)$.

This completes the proof in case 2.

Case 3. $p \in \Delta \setminus \Delta^\circ$ and $(\forall n) p^n \in \Delta \setminus \Delta^\circ$.

- Then, $\forall n, q^n \cdot p^n = 0$
- Therefore,

$$q \cdot p = \left(\lim_{n \rightarrow \infty} q^n \right) \cdot \left(\lim_{n \rightarrow \infty} p^n \right) = \lim_{n \rightarrow \infty} q^n \cdot p^n = 0$$

- Then $q \in \varphi(p)$

This completes case 3 and thus, φ has a closed graph.

By Kakutani's Fixed Point Theorem, $\exists p^* \in \Delta : p^* \in \varphi(p^*)$.

This implies, $p^* \in \Delta^\circ \wedge z(p^*) = 0$.

QED.

Existence in pure exchange economies

Corollary 1 (Exchange economies) In a pure exchange economy where consumption sets are the positive orthant, with complete, transitive, continuous, strictly convex, and strongly monotone preferences, and with strictly positive aggregate endowment (i.e. $\bar{\omega} \gg 0$), there is a Walrasian equilibrium.

Proof of corollary

Define $z : \Delta^\circ \rightarrow \mathbb{R}^L$ by $z(p) = \sum_{i=1}^I D_i(p) - \bar{\omega}$.

Verify z satisfies hypotheses of DGKN Lemma.

- z is a continuous function defined on Δ° .
- Walras Law holds because of strong monotonicity.
- Boundedness below: $D_i \geq 0$, then $D_i - \omega_i \geq -\omega_i$. Thus $z_i \geq -\omega_i$ and $z \geq -\bar{\omega}$.
- Boundary condition: Check.

Since z satisfies the hypotheses of DGKN Lemma, $\exists p^* \in \Delta^\circ$ such that $z(p^*) = 0$. Let $x_i^* = D_i(p^*)$. Then (p^*, x^*) is a WE.

Remarks

- Without LNS (and thus without monotonicity), it can be shown that $\exists p^* \in \Delta$, x_i^* is in the quasi-demand of agent i , and $\sum_{i=1}^I x_i^* \leq \bar{\omega}$.

Thus, quasi-Walrasian equilibrium exists. For some goods, demand may be less than supply.

If in addition $(\forall i) \omega_i \gg 0$, then $p^* \cdot \omega_i > 0$, $x_i^* \in D_i(p^*)$, and $\sum_{i=1}^I x_i^* \leq \bar{\omega}$.

- If one agent, say $i = 1$, has strongly monotone preferences and $\omega_1 \gg 0$, then $p^* \gg 0$ and then $x_i^* \in D_i(p^*)$ for every i and $\sum_{i=1}^I x_i^* \leq \bar{\omega}$.

If in addition all agents exhibit LNS, then $\sum_{i=1}^I x_i^* = \bar{\omega}$.

More remarks

- LNS does not imply that the allocation is exact; some prices may be zero.
- Strict convexity can be replaced by weak convexity: Demand becomes a correspondence.

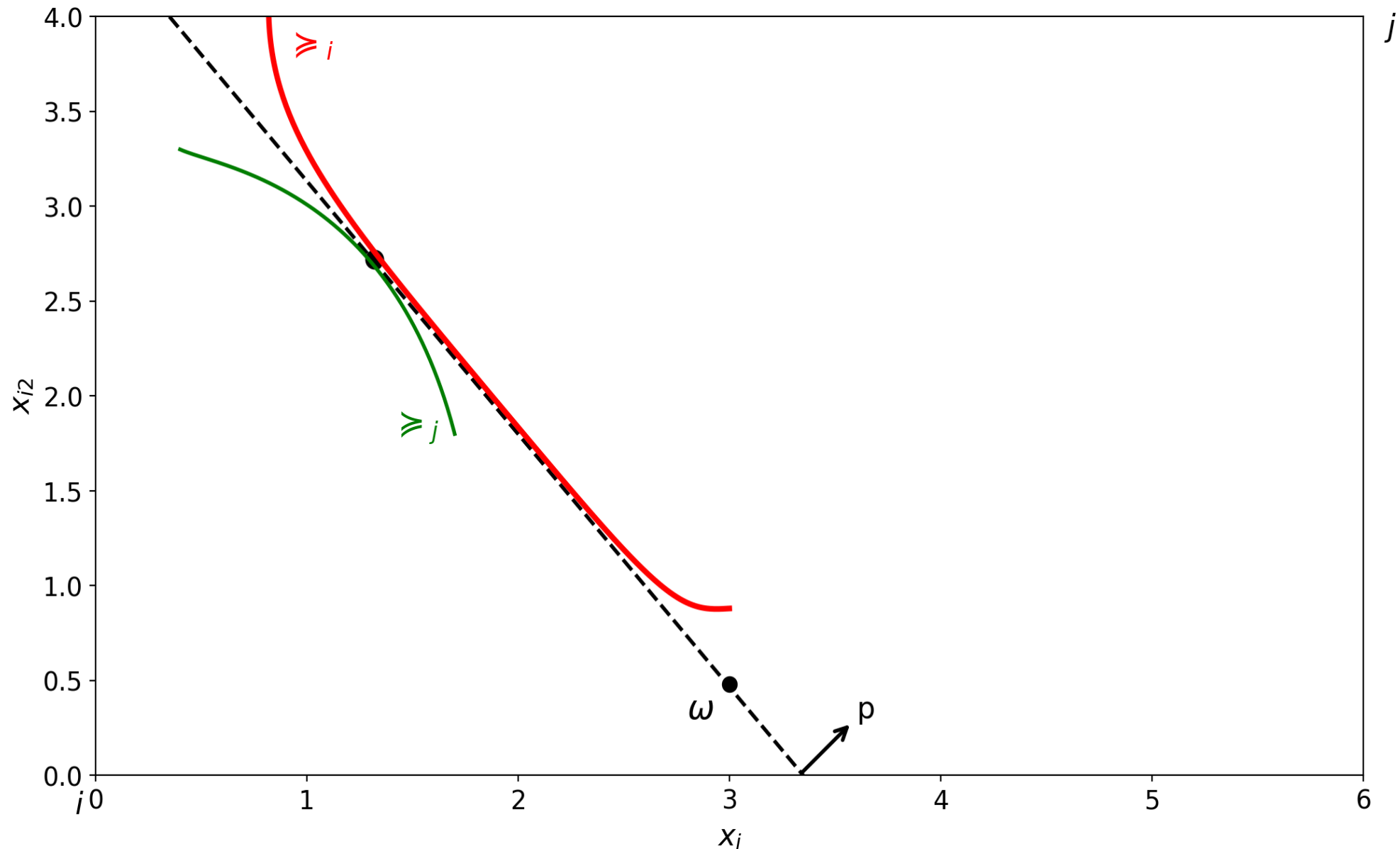
No decentralization property.

- With nonconvexities WE may not exist.

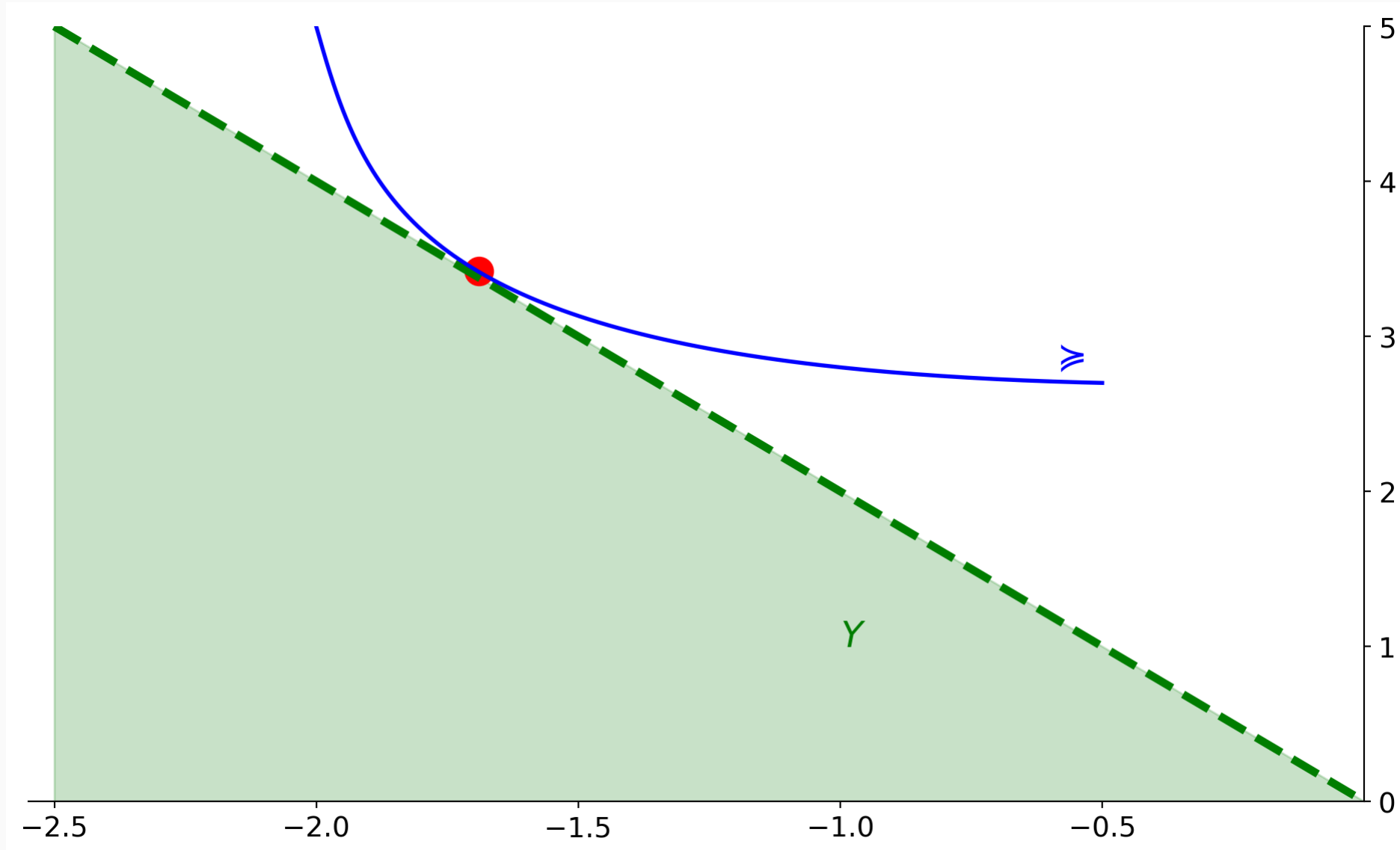
With nonconvexities there are approximate WE in large economies, i.e. economies with many relatively small agents.

No WE because of lack of income.

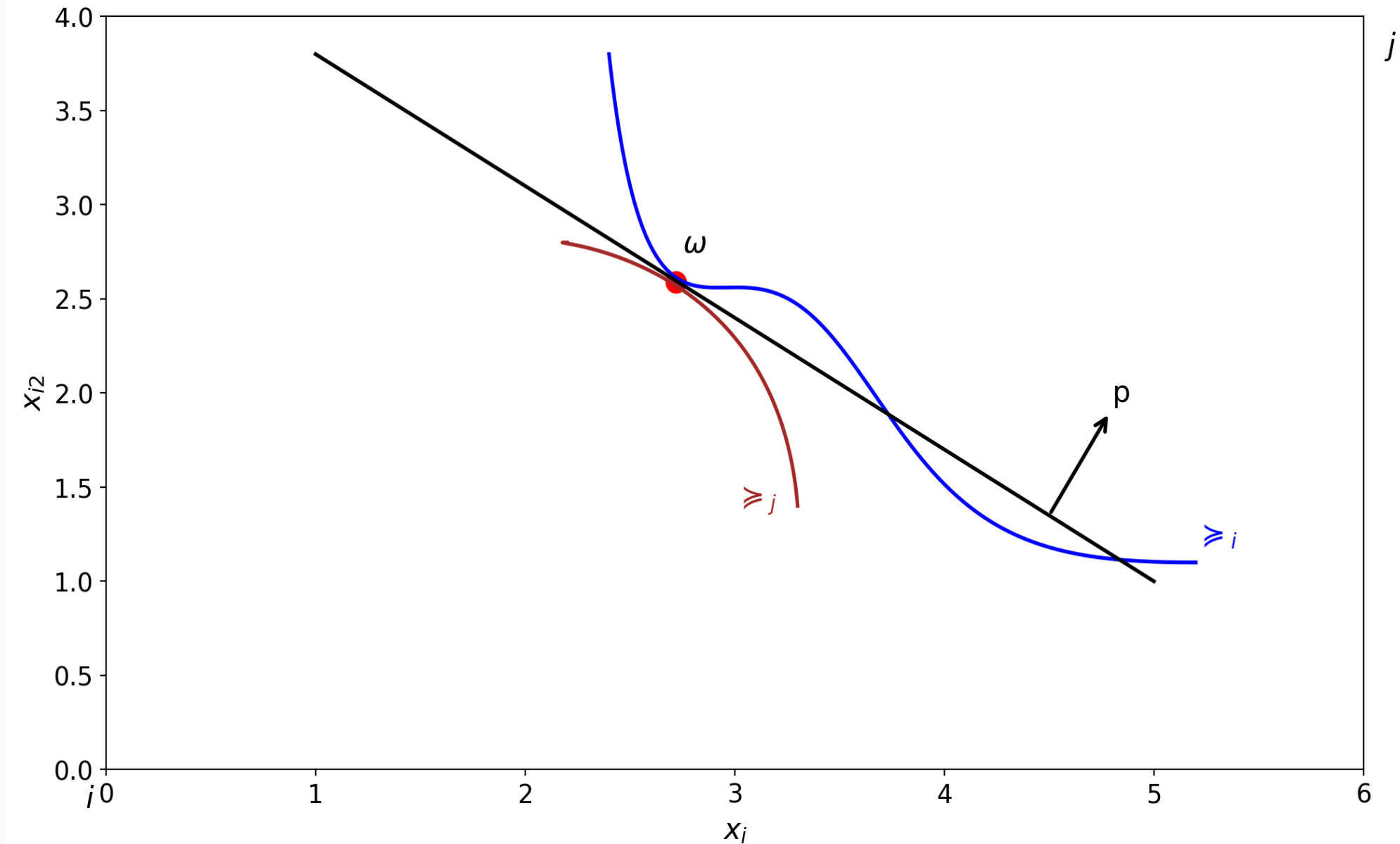
Weak convexity; lack of decentralization



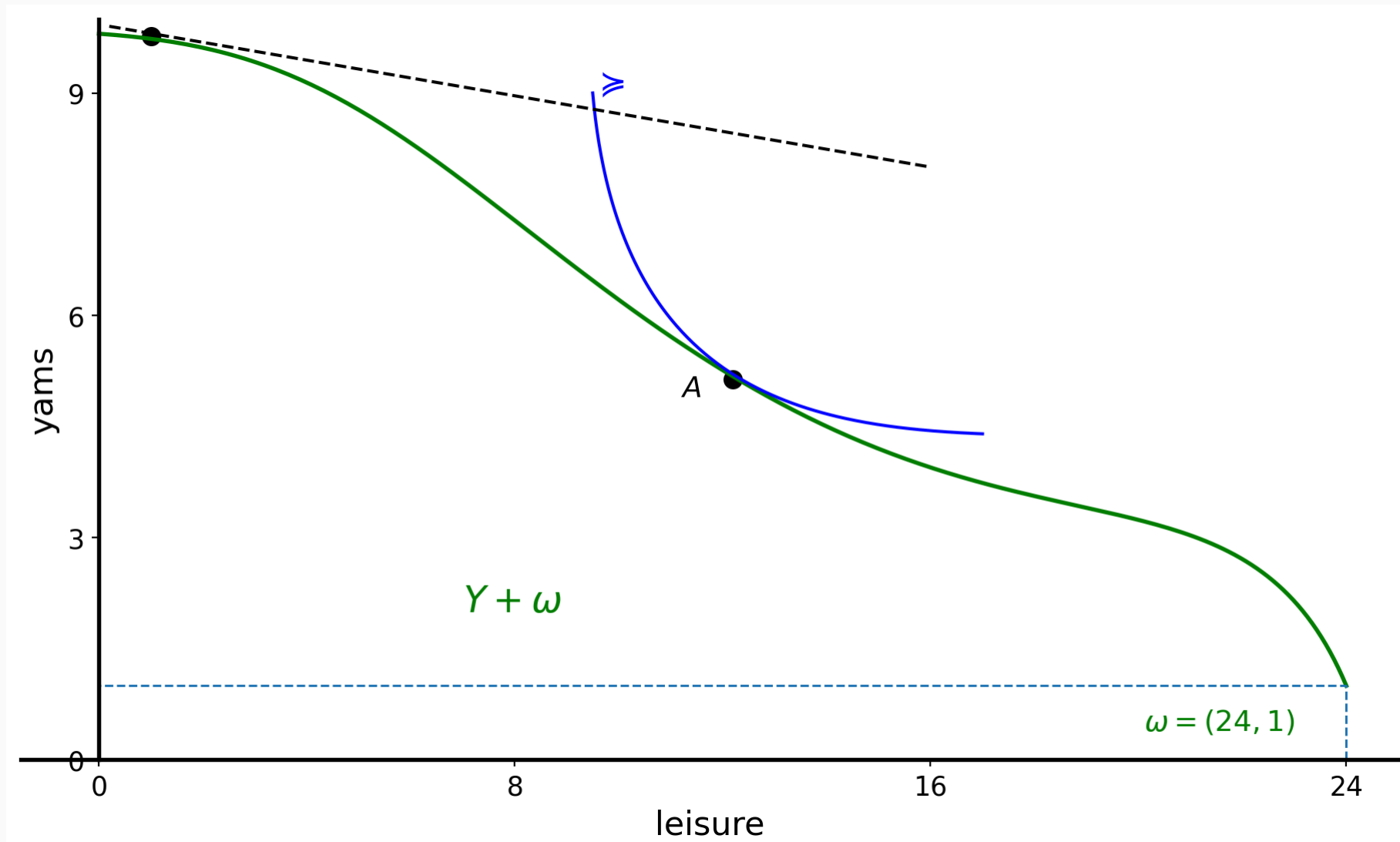
CRS and lack of decentralization



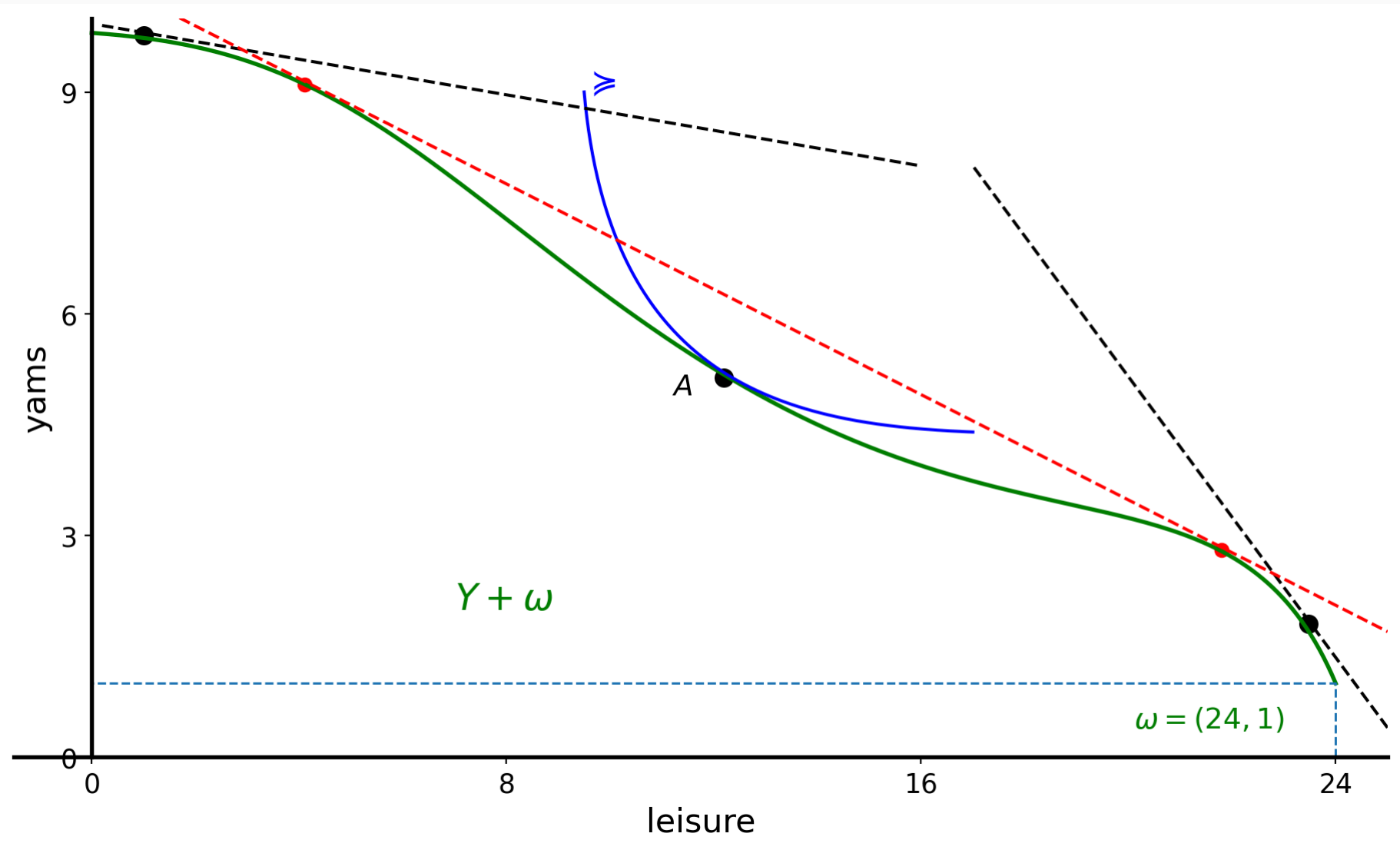
Failure of convexity



Failure of convexity in production



Failure of convexity in production



Non-convexities: approximate equilibria

References

Mas-Colell, Andreu, Michael Whinston, and Jerry Green. 1995. *Microeconomic Theory*. New York, Oxford: Oxford University Press.